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Real Analysis II

Past Papers Solution:

Q:1 Find first four nonzero terms in the Maclaurin series.

a) $e^x \sin x$

Solution:

We know that maclaurin series of $e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \cdots$

and $\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} + \cdots$

So

$$e^x \sin x = \left(1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \cdots\right)\left(x - \frac{x^3}{6} + \frac{x^5}{120} + \cdots\right)$$

$$= x - \frac{x^3}{6} + \frac{x^5}{120} + x^2 - \frac{x^4}{6} + \frac{x^6}{120} + \frac{x^3}{2} - \frac{x^5}{12}$$

$$+ \frac{x^7}{240} + \frac{x^4}{6} - \frac{x^6}{36} + \frac{x^8}{720} + \cdots$$

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2)

$= x + x^2 + \left(-\frac{1}{6} + \frac{1}{2}\right)x^3 + \left(-\frac{1}{12} + \frac{1}{120}\right)x^5 + \cdots$

$= x + x^2 - \frac{2}{6}x^3 - \frac{9}{120}x^5 + \cdots$

$= x + x^2 - \frac{x^3}{3} - \frac{3}{40}x^5 + \cdots$

6) $\frac{e^{-x}}{1+x^2}$

Solution:

Maclaurin series:

$e^{-x} = 1 - x + \frac{x^2}{2} - \frac{x^3}{6} + \cdots$

$\frac{1}{1+x^2} \Rightarrow (1+x^2)^{-1} = 1 - x^2 + \cdots$

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$= x + x^2 - \frac{x^3}{3} - \frac{3}{140}x^5 + \cdots$

b) $\frac{e^{-x}}{1+x^2}$

Solution:

Maclaurin series:

$e^{-x} = 1 - x + \frac{x^2}{2} - \frac{x^3}{6} + \cdots$

$\frac{1}{1+x^2} \Rightarrow (1+x^2)^{-1} = 1 - x^2 + \cdots$

Now

$e^{-x}(1+x^2)^{-1} = \left(1 - x + \frac{x^2}{2} - \frac{x^3}{6} + \cdots\right)\left(1 - x^2 + \cdots\right)$

$= 1 - x^2 - x + x^3 + \frac{x^2}{2} - \frac{x^4}{2} - \frac{x^3}{6} + \frac{x^5}{6} + \cdots$

$= 1 - x + \left(-1 + \frac{1}{2}\right)x^2 + \left(1 - \frac{1}{6}\right)x^3 + \cdots$

$= 1 - x - \frac{1}{2}x^2 + \frac{5}{6}x^3 + \cdots$

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3)

c) $\frac{\cos x}{1+x^6}$

Solution:

We know that Maclaurin series of

$\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720} + \cdots$

$\frac{1}{1+x^6} = (1+x^6)^{-1} = 1 - x^6 + \cdots$

Now

$\frac{\cos x}{1+x^6} = \left(1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720} + \cdots\right)(1 - x^6)$

$= 1 - x^6 - \frac{x^2}{2} + \frac{x^8}{2} + \frac{x^4}{24} - \frac{x^{10}}{24} - \frac{x^6}{720} + \frac{x^{12}}{720}$

$= 1 - \frac{x^2}{2} + \frac{x^4}{24} + \left(-1 - \frac{1}{720}\right)x^6 + \cdots$

$= 1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{721}{720}x^6 + \cdots$

d) $\sin x(\log(1+x))$

Solution:

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4)

We know that Maclaurin series is

$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots$$

We have

$$f(x) = \log(x+1)$$

$$f'(x) = \frac{1}{1+x}$$

$$f''(x) = \frac{d}{dx}\left(\frac{1}{1+x}\right)$$

$$= -1(1+x)^{-2} \Rightarrow -\frac{1}{(1+x)^2}$$

$$f'''(x) = \frac{d}{dx}\left(\frac{-1}{(1+x)^2}\right)$$

$$= -\left(-2(1+x)^{-3}\right) \Rightarrow \frac{2}{(1+x)^3}$$

$$f^{iv}(x) = \frac{-6}{(1+x)^4}$$

Now:

$$f'(0) = \frac{1}{1+0} = 1$$

$$f''(0) = -\frac{1}{(1+0)^2} = -1$$

$$f'''(0) = \frac{2}{(1+0)^3} = 2$$

$$f^{iv}(0) = -\frac{6}{(1+0)^4} = -6$$

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5)

Now put these values in Maclaurin Series

$$f(x) = 0 + 1.x - \frac{1}{2}x^2 + \frac{2}{6}x^3 - \frac{6}{24}x^4 + \cdots$$

$$= 0 + x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} + \cdots$$

So

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} + \cdots$$

and

$$\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} + \cdots$$

$$(\sin x)\log(1+x) = \left(x - \frac{x^3}{6} + \frac{x^5}{120} + \cdots\right)\left(x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \cdots\right)$$

$$= x^2 - \frac{x^3}{2} + \frac{x^4}{3} - \frac{x^5}{4} + \frac{x^6}{5} - \frac{x^4}{6} + \frac{x^5}{12} - \frac{x^6}{18} + \frac{x^7}{24} - \frac{x^8}{30}$$

$$+ \frac{x^6}{120} - \frac{x^7}{240} + \frac{x^8}{360} - \frac{x^9}{480} + \frac{x^{10}}{600} + \cdots$$

$$= x^2 - \frac{x^3}{2} + \left(\frac{1}{3} - \frac{1}{6}\right)x^4 + \left(-\frac{1}{4} + \frac{1}{12}\right)x^5 + \cdots$$

$$= x^2 - \frac{x^3}{2} + \frac{1}{6}x^4 - \frac{1}{6}x^5 + \cdots$$

6)

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6) Most Important

Q:2 Drive the identity
$2\sin x\cos x = \sin 2x$
from Maclaurin series
for $\sin x$, $\cos x$, $\sin 2x$

Proof:

L.H.S

$$2\sin x\cos x = 2\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots\right)\left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots\right)$$

$$= 2\left(x - \frac{x^3}{2!} + \frac{x^5}{4!} - \frac{x^3}{3!} + \frac{x^5}{3!2!} - \frac{x^7}{3!4!} + \frac{x^5}{5!} - \frac{x^7}{5!2!} + \frac{x^9}{5!4!} + \cdots\right)$$

$$= 2\left(x - \frac{x^3}{2} - \frac{x^3}{6} + \frac{x^5}{24} + \frac{x^5}{12} + \frac{x^5}{120}\right) \quad \text{Neglecting } x^7, x^9$$

$$= 2\left(x + \left(-\frac{1}{2} - \frac{1}{6}\right)x^3 + \left(\frac{1}{24} + \frac{1}{12} + \frac{1}{120}\right)x^5 + \cdots\right)$$

$$= 2\left(x - \frac{2}{3}x^3 + \frac{2}{15}x^5 + \cdots\right)$$

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7)

$= 2x - \frac{1}{3}x^3 + \frac{1}{15}x^5 + \cdots$

R.H.S

$\sin 2x$

Put $x = 2x$

$\sin 2x = \left(2x - \frac{(2x)^3}{3!} + \frac{(2x)^5}{5!} + \cdots\right)$

$= 2x - \frac{8x^3}{6} + \frac{32x^5}{120} + \cdots$

$= 2x - \frac{4}{3}x^3 + \frac{4}{15}x^5 + \cdots$

L.H.S = R.H.S

Proved

8)

Most Imp

Q:3 Starting from Maclaurin series $\log(1-x) = -\sum_{n=0}^{\infty} \frac{x^{n+1}}{n+1}$

$|x| < 1$ Show by Abel's theorem $\sum_{n=0}^{\infty} \frac{1}{(n+1)(n+2)}$

Solution:

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8)

back b Pg:35

$$f(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^n$$

then

$$\int_{x_0}^{x_1} f(x)\,dx = \sum_{n=0}^{\infty} \frac{a_n}{n+1} \left[(x_1 - x_0)^{n+1} - (x_0 - x_0)^{n+1}\right]$$

we

$$(x-1)\log(1-x) - x = -\sum_{n=0}^{\infty} \frac{x^{n+1}}{(n+1)(n+2)} \quad |x| < 1$$

Since $\sum_{n=0}^{\infty} \frac{1}{(n+1)(n+2)} < \infty$.

Abel's theorem implies that

$$\sum_{n=0}^{\infty} \frac{1}{(n+1)(n+2)} = -\lim_{x \to 0^-} \left((x-1)\log(1-x) - x\right) = 1$$

9.4 Prove if $\sum a_n$ is bounded

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WC

$$(x-1)\log(1-x)-x = -\sum_{n=0}^{\infty} \frac{x^{n+1}}{(n+1)(n+2)} \quad |x|<1$$

Since $\sum_{n=0}^{\infty} \frac{1}{(n+1)(n+2)} < \infty$.

Abel's theorem implies that

$$\sum_{n=0}^{\infty} \frac{1}{(n+1)(n+2)} = -\lim_{x \to 0^-} \left((x-1)\log(1-x)-x\right) = 1$$

Q:4 Prove if $(a_n r^n)$ is bounded and $|x_1 - x_0| < r$ then $\sum a_n (x_1 - x_0)^n < r$ converges.

Solution:

If $|a_n r^n| \leq M$, then

$|a_n (x_1 - x_0)^n| \leq M\rho^n$ where $\rho = |x_1 - x_0|/r < 1$

and

$\sum |a_n (x_1 - x_0)^n| < \infty$ because $\sum \rho^n < \infty$

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9)

Q:5 Show that the series $\sum_{n=0}^{\infty} \frac{\sin nx}{n^p}$ is uniformly convergent for all real values of x and $P>1$

Solution:

Since

$$\left|\frac{\sin nx}{n^p}\right| \leq \frac{1}{n^p}$$

and $\sum 1/n^p$, δ if $P>1$ the comparison test implies that

$$\sum \left|\frac{\sin nx}{n^p}\right|, \delta, \quad P>1$$

So the series converges absolutely if $P>1$

10)

Q:6 $|X+Y| \leq |X|+|Y|$

Proof:

By definition

$$|X+Y|^2 = \sum_{i=1}^{n} (x_i + y_i)^2$$

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10)

$$= \sum_{i=1}^{n} x_i^2 + 2\sum_{i=1}^{n} x_i y_i + \sum_{i=1}^{n} y_i^2$$

$$= |X|^2 + 2(X \cdot Y) + |Y|^2$$

$$\leq |X|^2 + 2|X||Y| + |Y|^2$$

$$= (|X| + |Y|)^2 \quad \text{by Schwarz's inequality}$$

Hence

$$|X+Y|^2 \leq (|X| + |Y|)^2$$

By taking squer root

$$|X+Y| \leq |X| + |Y|$$

Proved

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